CostVision
AI COST INTELLIGENCE

THE PROBLEM

What it costs us to cost a part today

Every part costed in CAPEE is typed in by hand. Here is that job, and here is what it is worth in your own numbers.

HOW A PART GETS COSTED TODAY

The engineer opens the 3D model, reads the drawing, and types the numbers into CAPEE by hand.

Boxes to fill, depending on the part type:
machined 12 · casting 30 · moulding 39 · pressing 69

WHY IT MATTERS TWICE

It takes time, and the time scales with the number of parts we want costed. That is the obvious cost.

Two engineers costing the same part do not type the same numbers, so one part gets two answers.

What it is worth: five numbers, four of them yours

	What it is	JLR's number	Where it comes from
A	Minutes to cost one part in CAPEE today	______	Time a handful of parts. One part type is enough to start
B	Of those, the minutes spent finding and typing the input values	______	The same exercise, timed separately
C	Parts costed through CAPEE in a year	______	CAPEE's own records
D	Parts in a typical basket that never get costed individually	______	Programme data. This is what Option 2 goes after
E	Share of the input boxes the engine fills on its own	______	We measure this in the pilot, per part type

Hours spent typing today
B × C ÷ 60

Hours freed a year
B × E × C ÷ 60

Extra parts that time would cost
B × E × C ÷ A

OPTION 1 · THE PILOT

Option 1: fill CAPEE from the 3D model

CAPEE keeps doing the costing. The engineer stops typing the input and starts checking it. No AI anywhere in this option.

Attach the 3D model

STEP, IGES or STL, from the CAPEE screen

The engine measures it

On a JLR server, with no internet connection

Rules fill the values

Fixed rules. Same part, same numbers, always

The engineer confirms

Only what a solid model cannot show

CAPEE costs the part

Exactly as it does now. It just gets the numbers

WHAT THE 3D MODEL GIVES, WITH NOTHING TYPED

Volume, weight, bounding box, surface area.
Wall thickness, holes, pockets, bosses, bend count, gear teeth, draft, machined face count.

165 costing rules turn those measurements into form values. 149 write into a named box.

WHAT THE ENGINEER STILL SUPPLIES

Material family. A solid model cannot tell steel from aluminium, and the same shape differs threefold in weight. It can come from the part master instead.

Tolerance class, finish, heat treatment, coating. The tool asks. It never guesses and never pre-ticks.

What it is built from, and what it does not use

The measuring engine is Python with the OpenCASCADE geometry kernel, the open engineering kernel behind FreeCAD and many commercial CAD tools. The rules that turn measurements into form values are ordinary code, not a model. There is no AI in this option and no outbound connection: the 3D file is read on a JLR server and never leaves it.

OPTION 2 · THE CAPABILITY

Option 2: cost the whole basket in one run

This is not a bigger version of Option 1. It is something we cannot do at all today.

WHAT WE DO TODAY

We cost the parts we have engineer-hours for.

The rest of a basket gets sampled, estimated by analogy, or taken from the last supplier quote.

WHAT OPTION 2 DOES

A list of parts and their 3D models goes in. Every one is measured, costed and reported in one run.

An engineer answers only what the geometry cannot decide — once, applied to every part it affects.

How long a run takes. The 40-part row was measured on real production parts; the rest carries that rate forward

A run of	Measured	Basis
One part, warm	**0.9 to 3.1 seconds**	Five real production parts, 0.6 MB to 3.0 MB. Bigger file, longer
40 parts	**31 seconds**	Measured. Two workers running in parallel, nothing reused from cache
500 parts	**about 6.5 minutes**	That measured rate carried forward. Roughly half of it on four workers

MUST BE BUILT					
	A history of rate changes	An automatic record of every costing	Region, volume and gear on the run	The safety checks on that route	The run itself

The costing itself needs no AI here either

The engine and the rules are the same ones Option 1 uses, and they make no model call. An agent is one way to marshal a run; taking the part type and material family off the part master is the other, and that is the route to take while AI is not approved. What does not exist today is the run itself, and the five things above.

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THE DECISION

The two options, and what we are asking for

	OPTION 1 · input into CAPEE	OPTION 2 · bulk costing
What it changes	**The engineer stops typing the input**	**We can cost a whole basket, not a sample**
Where the costing happens	**CAPEE, same as now**	CostVision engine, beside CAPEE
Does it use AI	**No. 3D model and fixed rules only**	**Not for the costing. The run can be driven off the part master**
Does anyone learn a new tool	**No, the same CAPEE screens**	A reviewer does, for the results
What it needs from JLR	A Linux server, the CAPEE connection, parts with prices we paid	Our own rate book, plus the five items built
Can we stop part way	**Yes, CAPEE costs the same way either way**	**Yes, it runs alongside**
Is it audit-ready today	**Yes, CAPEE keeps the record**	**No, until rate history and a costing record exist**

What we are asking for today: approve Option 1 as a pilot

One part type, 30 to 50 parts we have already bought, inside CAPEE, costing our own engineering time. It gives us the two numbers nobody can state today: how close the tool gets to a price we actually paid, and how much of the input it fills on its own. Option 2 needs those numbers first.

The one thing to know before you decide

CostVision has never been checked against a price JLR has actually paid. Not one part. That is exactly what the pilot settles, and until it is done nobody can tell you how accurate this is.